

MDM2 : Illuminati Ltd. - Modelling images of particle flow

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Dear Consultants,

To help our design programme move forward, Illuminati Ltd. needs your help. We design equipment for measuring fluid flows. Two of the most important things we want to know about a flow are its velocity field and density field, but velocities are notoriously difficult to measure with satisfactory accuracy, and density fields are only slightly easier. In fact both are challenging inference (reverse-engineering) problems.

The standard way of providing raw data from which a velocity field can be inferred is to illuminate the flow using a laser beam that is optically spread to form a sheet of light on some central plane within the domain. To produce a signal with good optical contrast, reflective spherical particles are suspended in the fluid (usually water or air) so that their distribution is reasonably uniform. Simultaneously, we dope some regions of the fluid with a fluorescent dye so that in situations where buoyancy due to contrasts in density is a major driver of the flow, we can track the advection of the density field. A video camera acquires a long sequence of images, each separated by a known time interval Δt .

Provided the displacement $\Delta \mathbf{r}$ undergone by an individual particle at initial position $\mathbf{r}$ as a result of being carried by the flow can be identified, then the velocity vector for that particle is simply $\mathbf{v} \approx \frac{\Delta \mathbf{r}}{\Delta t}$. The challenge arises from having to decide how to identify a particle in one image frame of the video sequence with a particle in the next frame, when they all look the same. There is no easy answer to this question. The standard - not very good - approach is to treat groups of particles in one frame as a pattern and seek to match this pattern in the next frame. This method is known as 'Particle Image Velocimetry', and involves calculating cross-correlations between a small patch in one frame and the same patch in the subsequent frame in the image sequence to determine one displacement vector in the field. False correlations are an inevitable hazard of this method.

The density estimate is inferred from the light intensity produced by the fluorescent dye. As dyed fluid mixes with undyed fluid, the concentration of dye in the resulting mixture is a linear combination by volume. However the corresponding light intensity seen by the video camera is not a straightforward function of dye concentration, since laser light emanates from a point and spreads, reducing intensity with distance, and furthermore, decays in intensity along a

light ray due to absorption by the dye itself (governed by the Lambert-Beer law).

At Illuminati Ltd. we would like to jettison the standard Particle Image Velocimetry method and instead develop a statistical inference engine that infers the coupled evolution of a particle position field and a density field, given a sequence of images of a flow. For each individual image there exists a map $\mathbf{A}$, say, that takes an unknown state vector of particle position coordinates and dye concentrations across all locations in the domain, $\mathbf{x}$, to a corresponding pixel image $\mathbf{b}$ as follows:

$$\mathbf{A}(\mathbf{x}) = \mathbf{b} \quad (1)$$

Unfortunately, because of all the above complications of how the image is produced, $\mathbf{A}$ is unlikely to be linear and unlikely to be invertible. The best we can hope to achieve even in ideal conditions is a minimisation of

$$\|\mathbf{A}(\mathbf{x}) - \mathbf{b}\|_2. \quad (2)$$

Useful additional information about particle and dye movement may be gained from looking at pairs and triplets of images, rather than only considering images in isolation, so we may wish to redefine $\mathbf{b}$ to be the stacked vector containing all images in the sequence, $\mathbf{x}$ as the corresponding stacked vector of states and $\mathbf{A}$ as a map over the entire video sequence. Either perspective may prove useful.

At Illuminati Ltd. we believe a sensible approach to this problem is to build a comprehensive forward model for the mapping $\mathbf{A}$, and separately build a method for solving (2) that is robust in the context of the data. The forward model itself may be decomposed into 1) a simple Lagrangian evolution model for the dye and particles that conserves mass, momentum and energy over a few Δt (ie. before particle trajectories get too tangled), and 2) an optical model for how particles and dye together generate an image in the camera sensor plane.

As a useful placeholder for a fully-fledged fluid evolution model, the following velocity field may be used to evolve particles, storing coordinates in a random order at increments of Δt :

$$u(x, z, t) = \begin{cases} Re(ikAe^{-kz}e^{ikx}e^{\gamma t}), & z > 0 \\ Re(ikAe^{kz}e^{ikx}e^{\gamma t}), & z < 0 \end{cases} \quad (3)$$

$$w(x, z, t) = \begin{cases} Re(-kAe^{-kz}e^{ikx}e^{\gamma t}), & z > 0 \\ Re(kAe^{kz}e^{ikx}e^{\gamma t}), & z < 0 \end{cases} \quad (4)$$

where Re represents the real part of a complex variable, A is a scale parameter, k is a horizontal wavenumber of a displaced interface around $z = 0$, γ is a growth rate, and the remaining symbols take their usual meaning. Density (and dye concentration) above the interface is taken to be greater than density below. The domain under consideration should be cuboidal and the ideal initial distribution of particles is random, though for practical reasons they may cluster around $z = 0$ in real experimental data.

Real data from experiments using Illuminati equipment will be made available, if the need arises, to test a fully functional inference engine. Note that in

real systems, the flow often contains features at scales below that of the resolution provided by the particles. Appreciable discrepancy may then occur between a model at the particle scale and reality, and your method should account for this. Furthermore, in real data, the full 3D field is unseen except for a single planar slice, and particles appear and disappear between frames because their velocity may have a component out of the illuminated plane. Illuminati Ltd. thank you in advance for your time and effort and look forward to your report.